

# Integration Testing

Validation of interactions among components, services, databases, queues and external systems.

## Interfaces

Validate contracts, formats, status, timeouts, retries, authentication, idempotency and error handling.

## Data

Confirm mappings, transformations, persistence, consistency and effects across systems.

## Dependencies

Use real environments when needed and mocks/stubs when they help isolate risk, understanding limitations.

## Observability

Logs, traces, correlation IDs and metrics help locate integration failures.

## When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

## How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

## QA tip

Many production defects live at system boundaries rather than inside isolated components.